Balanced transport at the egalitarian core: a complete boundary phase diagram

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Abstract

For a transferable-utility game, exact partition at an allocation means that every cumulative union of its richest payoff strata is tight. Both this condition and its Lorenz consequence are due to earlier work. We determine exactly when the condition can fail at the leximin core allocation of a singleton-normalized exact game. If a strict payoff boundary has m cumulative poorer players and h richer players, then a slack richer suffix exists if and only if

$$m \geq m_{\min}(h), \qquad m_{\min}(1) = 4, \quad m_{\min}(2) = m_{\min}(3) = 3, \quad m_{\min}(h) = 2 \ (h \geq 4).$$

The lower bounds are structural. A normal-cone theorem turns leximin optimality into one simultaneous strict cover by tight coalitions. Projecting the cover to the richer side and applying a convex defect functional shows more generally that every failure selects one lifted minimal balanced packet with a positive rooted semi-balanced obstruction; mixtures are unnecessary. The resulting certificate uses at most $|N|-1$ positive coalitions, and its slack budget forces some one-player extension of the richer suffix to be slack as well. One poorer player cannot support the required opposing signs. A unique richest player needs four poorer players by a fractional sacrifice budget. With two poorer players, every failure needs a proper active hyperedge of size at least three, hence at least four richer players. Three explicit dimension-free lower-envelope families attain every remaining pair. The familiar four-player guarantee and sharp five-player failure are immediate corollaries. The argument enumerates no games and no payoff-stratum patterns; its only case analyses are over the shapes of a protector trace and over the components of a rank-two vertex.

1 Introduction and prior-art boundary

Let (N, v) be a transferable-utility game and

$$\mathrm{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for every } S \subseteq N\}.$$

The game is *exact* if every coalition inequality is attained by some core allocation. Write $\phi = \mathrm{Lmin}(v)$ for the unique core allocation whose increasing rearrangement is lexicographically maximal. At a strict payoff boundary of ϕ, let L be the cumulative poorer side and let $H = N \setminus L$ be the richer suffix.

The exact-partition condition and its implications for Dutta–Ray and Lorenz selection are prior art (Llerena and Mauri, 2017; Sánchez-Soriano et al., 2010). Qualitative tight-coalition descriptions of the egalitarian selections are likewise established: Arin and Iñarra (2001) characterize the egalitarian core by the impossibility of a rich-to-poor transfer inside the core, which is exactly the existence of a tight coalition containing the richer player and omitting the poorer one, and they place Lmin in that set; Arin et al. (2003) characterize Lmax on large-core games; see also

Mauri (2018). Exact balancedness and semi-balanced facet systems are also established (Csóka et al., 2011; Lohmann et al., 2012; Studený and Kratochvíl, 2022). In particular, the latter authors already obtain purely minimal semi-balanced systems from minimal balanced systems by complementing the exceptional coalition. Our candidate contribution is not that full-set correspondence, but the Lmin-selected normal cover and its projection to one minimal balanced packet on the richer suffix, together with the sharp two-parameter classification and realizing families. The closest published low-dimensional result known to us is that exact games with at most four players have large cores (Biswas et al., 1999) and that large-core games are egalitarian stable, with PES = Lmax on that domain (Dietzenbacher, 2021); that chain does not itself assert exact partition at Lmin.

The main theorem is stated now. All games below are singleton-normalized, meaning $v(\{i\}) = 0$ for every player.

Theorem 1 (boundary phase diagram)**.** *At a strict payoff boundary of $\phi = \mathrm{Lmin}(v)$ in a singleton-normalized exact game, put $m = |L|$ and $h = |H|$. It is possible that $v(H) < \phi(H)$ exactly for*

$$m \geq m_{\min}(h), \qquad m_{\min}(h) = \begin{cases} 4, & h = 1, \\ 3, & h = 2 \text{ or } 3, \\ 2, & h \geq 4. \end{cases} \tag{1}$$

Every admitted pair is realized by an integral indecomposable exact game. The poorer side may contain several payoff strata.

The proof separates a general lower-bound mechanism from three universal constructions. This makes the four/five threshold a corollary rather than the theorem's organizing principle.

2 Strict covers and balanced-trace transport

For $\phi \in \mathrm{Core}(v)$ write

$$s_\phi(S) = \phi(S) - v(S).$$

A coalition is ϕ-tight when its slack is zero.

Lemma 2 (strict richer cover)**.** *Let $\phi = \mathrm{Lmin}(v)$ and let $N = L \,\dot\cup\, H$ be a strict payoff boundary. There are nonnegative weights on proper ϕ-tight coalitions whose incidence load is a common positive number on H and is strictly smaller at every player in L.*

Proof. Let $\mathcal{D}$ be the tangent cone of the core at ϕ. There is no $d \in \mathcal{D}$ with $d_i \geq 0$ for every $i \in L$ and one of these inequalities strict: a sufficiently short segment $\phi + td$ remains in the core, preserves the strict gap between L and H, and improves the first poorer payoff stratum on which d is nonzero.

Let $K \subseteq \mathbb{R}^L$ be the image of $\mathcal{D}$ under the coordinate projection. It is a polyhedral cone, being the linear image of a polyhedral cone, and the previous paragraph says $K \cap \mathbb{R}^L_+ = \{0\}$.

We claim the polar cone $K^\circ = \{w \in \mathbb{R}^L : \langle w, y\rangle \leq 0 \text{ for all } y \in K\}$ contains a *strictly* positive vector. Ordinary separation does not give this: two cones meeting only at the origin admit a separating $w \neq 0$ with $w \geq 0$, and such a w may vanish on some coordinates. Take polars instead. Polarity turns the intersection into a sum, so

$$\mathbb{R}^L = \{0\}^\circ = (K \cap \mathbb{R}^L_+)^\circ = \mathrm{cl}\big(K^\circ + (\mathbb{R}^L_+)^\circ\big) = \mathrm{cl}\big(K^\circ + \mathbb{R}^L_-\big).$$

Both K° and $\mathbb{R}^L_-$ are polyhedral, and a sum of finitely many polyhedral cones is polyhedral, hence closed; the closure operation is therefore vacuous and $K^\circ + \mathbb{R}^L_- = \mathbb{R}^L$. Write $\mathbf{1}_L = w + p$ with

$w \in K^\circ$ and $p \le 0$. Then $w = \mathbf{1}_L - p \ge \mathbf{1}_L > 0$. Thus $w_i > 0$ for every $i \in L$ and $\sum_{i\in L} w_i d_i \le 0$ for all $d \in \mathcal{D}$.

Extend w by zero on H; the displayed inequality is unchanged, so w lies in the polar of $\mathcal{D}$ in $\mathbb{R}^N$. Since $\mathcal{D} = \{d : d(N) = 0,\ d(T) \ge 0 \text{ for every proper } \phi\text{-tight } T\}$, that polar is generated by $\pm\mathbf{1}_N$ and by $-\mathbf{1}_T$, giving

$$w = \mu\mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T, \qquad \lambda_T \ge 0, \tag{NC}$$

where T ranges over proper ϕ-tight coalitions. Thus the weighted incidence load is μ on H and $\mu - w_i < \mu$ at $i \in L$. Nonnegativity of the load and positivity of w_L imply $\mu > 0$. Deleting coalitions disjoint from H preserves the richer load and only lowers poorer loads. □

Normalize the common rich load in Lemma 2 to one and remove coalitions disjoint from H. Project every remaining tight coalition to its nonempty trace on H. The projected weights lie in the exact-cover polytope

$$\mathcal{P}_H = \left\{ \beta \ge 0 : \sum_{\emptyset \ne R \subseteq H} \beta_R \mathbf{1}_R = \mathbf{1}_H \right\}. \tag{2}$$

Its vertices are the unique normalized weights of minimal balanced collections on H; this is standard balanced-collection geometry (Lohmann et al., 2012; Laplace Mermoud et al., 2025).

Theorem 3 (balanced-trace transport)**.** *Suppose $s = s_\phi(H) > 0$. Call a* lifted packet *a pair consisting of a vertex β of $\mathcal{P}_H$ together with a choice, for each trace R in its support, of one supported ϕ-tight coalition whose trace on H is R; its* poor-load vector *$r \in \mathbb{R}_+^L$ records the total weight of chosen coalitions containing each $i \in L$. Then there are a core point z, a displacement $\Delta = z - \phi$, and a finite convex mixture $\{(\theta_q, r^q)\}$ of lifted packets such that*

$$\Delta(H) = -s, \qquad \Delta(L) = s, \tag{3a}$$

$$\min_{i\in L} \Delta_i < 0 < \max_{i\in L} \Delta_i, \tag{3b}$$

$$-s + \sum_{i\in L} r_i^q \Delta_i \ge 0 \qquad \text{for every } q, \tag{3c}$$

$$\sum_q \theta_q r_i^q < 1 \qquad \text{for every } i \in L. \tag{3d}$$

Proof. Exactness supplies $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$, proving (3a). Let $c_i < 1$ be the poorer loads of a normalized cover from Lemma 2. Every supported coalition is tight at ϕ and feasible at z, hence

$$0 \le \sum_T \lambda_T \Delta(T) = -s + \sum_{i\in L} c_i \Delta_i. \tag{4}$$

If every poorer displacement were nonnegative, the right side would be strictly smaller than $-s + \Delta(L) = 0$. Since $\Delta(L) = s > 0$, both signs occur, proving (3b).

Let $\beta_R = \sum_{T\cap H = R} \lambda_T$ be the projected weights, so $\beta \in \mathcal{P}_H$, and decompose $\beta = \sum_p \theta_p \beta^p$ into vertices. Independently for each trace R, choose a preimage T with $T \cap H = R$ with conditional probability λ_T/β_R. A vertex together with such a section is a lifted packet, and the product measure assigns it the weight $\theta_p \prod_R (\lambda_{T_R}/\beta_R)$; index the resulting finite mixture by q.

Fix one lifted packet q, with vertex β^p and section σ. Each $\sigma(R)$ is tight at ϕ and feasible at z, so $\sum_R \beta_R^p \Delta(\sigma(R)) \ge 0$. A rich player j lies in $\sigma(R)$ exactly when $j \in R$, so its load is $\sum_{R\ni j} \beta_R^p = 1$, while a poor player i has load r_i^q by definition. The displayed sum therefore equals $\Delta(H) + \sum_{i\in L} r_i^q \Delta_i = -s + \sum_{i\in L} r_i^q \Delta_i$, proving (3c) for every packet, not merely on average.

For (3d), average r_i over the product measure. Since $\sum_p \theta_p \beta_R^p = \beta_R$ and the section is chosen independently per trace,

$$\sum_q \theta_q r_i^q = \sum_R \beta_R \cdot \frac{1}{\beta_R} \sum_{T \cap H = R,\ i \in T} \lambda_T = \sum_{T \ni i} \lambda_T = c_i < 1.$$

□

The mixture in Theorem 3 always contains one packet that already carries a positive rooted obstruction. This gives a general dimension-free reduction before the cardinality arguments begin.

Theorem 4 (one-packet reduction)**.** *At every slack strict boundary there is a lifted vertex of $\mathcal{P}_H$, with poorer load vector $r \geq 0$, such that, on putting*

$$b = \max_{i \in L} r_i, \qquad u_i = b - r_i, \qquad \delta = 1 + \sum_{i \in L} u_i - b,$$

one has $\delta > 0$ and

$$\sum_T \lambda_T \mathbf{1}_T + \sum_{i \in L} u_i \mathbf{1}_{H \cup \{i\}} - \delta \mathbf{1}_H = b \mathbf{1}_N. \tag{P}$$

Here the T are proper ϕ-tight coalitions and their traces form one minimal balanced collection on H. Moreover,

$$\delta s_\phi(H) \leq \sum_{i \in L} u_i s_\phi(H \cup \{i\}), \tag{PB}$$

so at least one u_i is positive. The trace collection is either the full-trace packet $\{H\}$ or consists entirely of proper subsets of H. The identity uses at most $|H|$ packet coalitions and at most $|L| - 1$ spokes, hence at most $|N| - 1$ positive coalitions.

Proof. Let $c_i = \sum_q \theta_q r_i^q < 1$ be the average loads in (3d). For a nonnegative vector a define

$$F(a) = 1 + (|L| - 1) \max_i a_i - \sum_i a_i.$$

This function is convex, and $F(c) \geq 1 - \max_i c_i > 0$. Jensen's inequality therefore gives a lifted vertex with $F(r) > 0$; this is exactly the displayed δ. Its rich load is one. In (P), poor player i has load $r_i + u_i = b$, while every rich player has load $1 + \sum_i u_i - \delta = b$.

The positive incidence columns at a vertex of $\mathcal{P}_H$ are linearly independent: otherwise a sufficiently small signed kernel perturbation would remain nonnegative in both directions. Thus the packet has at most $|H|$ traces. The pure lift uses one coalition per trace. Any i attaining the maximum b has $u_i = 0$, so at most $|L| - 1$ spokes are positive, proving the sparsity bound.

Choose $z \in \mathrm{Core}(v)$ exact on H. Evaluate (P) at z and at ϕ. Writing $\Delta = z - \phi$, the packet terms satisfy $\Delta(T) \geq 0$, the spokes satisfy $\Delta(H \cup \{i\}) \geq -s_\phi(H \cup \{i\})$, and $\Delta(H) = -s_\phi(H)$. The displacement form of (P) is therefore

$$0 = \sum_T \lambda_T \Delta(T) + \sum_i u_i \Delta(H \cup \{i\}) - \delta \Delta(H) \geq -\sum_i u_i s_\phi(H \cup \{i\}) + \delta s_\phi(H),$$

which is (PB). If every u_i were zero, (PB) would contradict $\delta s_\phi(H) > 0$. Finally, a minimal balanced collection containing H must be the singleton collection $\{H\}$; otherwise all its traces are proper. □

Thus every failure has an exterior full-trace packet or an internal minimal balanced-transport packet. The incidence identity is an established semi-balanced system with exceptional coalition H; the new point proposed here is that the Lmin normal certificate and the convex functional F select one minimal balanced packet on H from an arbitrary failure. This is different from the established complement-of-the-exceptional-set correspondence, which operates on the full player set.

Remark 5 (locating the packets in the published catalogue)**.** *Every system (P) is semi-balanced with exceptional coalition H in the sense of Studený and Kratochvíl (2022), and is minimal in every instance computed for this paper. Their electronic catalogue lists the permutational types of the* indecomposable *min-semi-balanced systems on at most six players, sixteen on five players and fifty-three on six; these are exactly the types that index facets of the cone of exact games. Indecomposability is here a property of the set system and is unrelated to direct-sum indecomposability of a game. Five of the five-player types and fifteen of the six-player types have a singleton exceptional coalition, which is necessary for an $h = 1$ obstruction. It is not sufficient: at $h = 1$ the exact-cover polytope (2) has the single vertex $\{H\}$ with weight one, so a system (P) has one lift, at most one positive coalition of size exceeding two, and all positive coefficients equal to one. Only three of the five singleton-exceptional five-player types have that form; the other two are never systems (P), at any boundary of any exact game on any number of players.*

Take the smallest realization of each of the three minimal pairs of the staircase (1). For the $(4, 1)$ member of Proposition 14 with richest player e, the tight proper coalitions at ϕ are ce, de, cde, $acde$, $bcde$, and every admissible packet coalition produces a catalogued type:

$$\begin{aligned} T = ce \text{ or } de: &\quad \mathbf{1}_{ae} + \mathbf{1}_{be} + \mathbf{1}_{ce} + \mathbf{1}_{de} - 3 \cdot \mathbf{1}_{e} = \mathbf{1}_N, \\ T = cde: &\quad \mathbf{1}_{ae} + \mathbf{1}_{be} + \mathbf{1}_{cde} - 2 \cdot \mathbf{1}_{e} = \mathbf{1}_N, \\ T = acde: &\quad \mathbf{1}_{acde} + \mathbf{1}_{be} - \mathbf{1}_{e} = \mathbf{1}_N. \end{aligned}$$

For the $(3, 2)$ member of Proposition 16, where $\phi = (1, 1, 2, 3, 3)$ and $H = de$, the only tight proper coalitions are ade and cde; both lifts give the full-trace packet, and the single resulting type is again catalogued:

$$\mathbf{1}_{ade} + \mathbf{1}_{bde} + \mathbf{1}_{cde} - 2 \cdot \mathbf{1}_{de} = \mathbf{1}_N.$$

For the $(2, 4)$ member of Proposition 15, where $\phi = (2, 2, 4, 4, 4, 4)$ and $H = cdef$, three types occur:

$$\begin{aligned} &\textit{catalogued:} && \mathbf{1}_{acdef} + \mathbf{1}_{bcdef} - \mathbf{1}_{cdef} = \mathbf{1}_N, \\ &\textit{not catalogued:} && \mathbf{1}_{ac} + \mathbf{1}_{def} + \mathbf{1}_{bcdef} - \mathbf{1}_{cdef} = \mathbf{1}_N, \\ &\textit{not catalogued:} && \mathbf{1}_{cef} + \mathbf{1}_{def} + \mathbf{1}_{acde} + \mathbf{1}_{acdf} + 2 \cdot \mathbf{1}_{bcdef} - 3 \cdot \mathbf{1}_{cdef} = 2 \cdot \mathbf{1}_N. \end{aligned}$$

The last two systems are min-semi-balanced but decomposable, by $E = \{c\}$ and by $E = \{c, d, e\}$; for the second system, for instance, $\mathbf{1}_{ac} + \mathbf{1}_{bcdef} - \mathbf{1}_{c} = \mathbf{1}_N$ is a combination with nonnegative coefficients inside the system and a negative one on E. Two points follow. First, the certificate of Theorem 4 is always a valid inequality for the exact cone but need not be a facet, so being catalogued is strictly stronger than being a system (P). In the instances computed, non-facet certificates arose only in the two-poorer family. Second, a single failure does not select a single class: which class appears depends on the tight coalition the cover supports. This is the concrete form of the multi-class phenomenon noted in Section 6. Decomposability is a statement about set systems alone; here $bcdef$ carries positive slack at ϕ, so it exhibits no sharper certificate at this boundary.

Two consequences are worth isolating. The first records what the certificate costs; the second is a propagation statement that needs no cardinality hypothesis.

Corollary 6 (linear-size witness and slack propagation)**.** *Let a strict payoff boundary of $\phi = \mathrm{Lmin}(v)$ have $s_\phi(H) > 0$. Then:*

(i) *the obstruction is exhibited by at most* $|N|-1$ *positive coalitions together with* H*, and by the rational data* λ, u, δ, b *of (P);*

(ii) $s_\phi(H) \le \dfrac{1}{\delta}\displaystyle\sum_{i\in L} u_i s_\phi(H\cup\{i\})$*;*

(iii) *consequently* $s_\phi(H\cup\{i\}) > 0$ *for at least one* $i \in L$*.*

Proof. Part (i) is the sparsity bound of Theorem 4. Part (ii) is (PB) divided by $\delta > 0$. For (iii), the right side of (ii) is positive because the left side is, and every u_i is nonnegative, so some product $u_i s_\phi(H\cup\{i\})$ is positive. □

Part (iii) says that a slack richer suffix is never isolated: the failure propagates to a one-player enlargement of H. Since the enlargement is still a proper coalition, iterating is not possible in general, and the statement is genuinely about the single step. Part (ii) quantifies this: the slack at H is financed, at exchange rate δ, by the slacks of at most $|L|-1$ single-player completions.

The first exclusion is immediate.

Corollary 7. *A slack boundary cannot have* $m = 1$*.*

Proof. Equation (3b) requires both a positive and a negative displacement on L. □

3 The two remaining lower bounds

3.1 A unique richest player

Assume $H = \{r\}$, write $\phi_r = \gamma$, and put $q_i = \phi_i < \gamma$ for $i \in L$. The transfer-blocking characterization of the egalitarian core (Arin and Iñarra, 2001) gives, for each $i \in L$, a tight protector

$$T_i = \{r\}\cup Q_i, \qquad \emptyset \ne Q_i \subseteq L\setminus\{i\}. \tag{5}$$

The nonemptiness follows because the normalized singleton $\{r\}$ is slack. Alternatively (5) is the case $H = \{r\}$ of Lemma 2: the cover has load μ at r and load $\mu - w_i < \mu$ at i, so the supported coalitions containing r but not i carry total weight $w_i > 0$ and at least one of them is positive. The argument below therefore does not depend on the cited characterization.

Lemma 8 (sacrifice budget)**.** *For weights* $\lambda_i \ge 0$ *satisfying*

$$c_j := \sum_{i: j\in Q_i} \lambda_i \le 1, \qquad \Lambda := \sum_i \lambda_i,$$

one has

$$\gamma(\Lambda - 1) \le \sum_{j\in L}(1-c_j)q_j. \tag{6}$$

Proof. Exactify $\{r\}$ at a core point z and put $d = z - \phi$. Summing the tight protector inequalities with the displayed weights, using efficiency and the singleton lower bounds $z_j \ge 0$, gives (6). Equivalently, complete the uncovered poorer loads by singleton coalitions; together with negative weight $\Lambda - 1$ on $\{r\}$, the incidence vector is exact-balanced, and the exact-balanced slack inequality is exactly (6). □

Proposition 9. *If* $h = 1$ *and the suffix is slack, then* $m \ge 4$*.*

Proof. For $m = 1$ no protector in (5) exists. For $m = 2$, both protector traces are the opposite singleton; weight both by one in (6), obtaining $\gamma \le 0$.

Let $m = 3$. If some $Q_i = \{j\}$, weight the protectors indexed by i and j by one. Their traces are disjoint and leave at most one player u uncovered, so (6) gives either $\gamma \le 0$ or $\gamma \le q_u < \gamma$. If no Q_i is a singleton, then $Q_i = L \setminus \{i\}$ for all i. Weight all three protectors by $1/2$; every poorer player has load one and (6) gives $\gamma/2 \le 0$. □

3.2 Exactly two poorer players

We first isolate the exact-face inequality used in the rounding argument.

Lemma 10 (exact-face overload)**.** *Let v be singleton-normalized and exact, let $\phi \in \mathrm{Core}(v)$, and let $D \subseteq H \subsetneq N$. Suppose nonnegative weights on proper ϕ-tight coalitions have load $1 + q$ on D, load one on $H \setminus D$, and loads $0 \le c_i \le 1$ on $i \in N \setminus H$, where $q > 0$. Then*

$$q s_\phi(D) \le \sum_{i \notin H} (1 - c_i)\phi_i. \tag{7}$$

Proof. Choose $x \in \mathrm{Core}(v)$ with $x(D) = v(D)$ and put $d = x - \phi$. Summing the supported tight-coalition inequalities at x gives

$$0 \le d(H) + q d(D) + \sum_{i \notin H} c_i d_i.$$

Efficiency replaces $d(H)$ by $-\sum_{i \notin H} d_i$. Hence

$$q s_\phi(D) = -q d(D) \le \sum_{i \notin H} (1 - c_i)(-d_i) \le \sum_{i \notin H} (1 - c_i)\phi_i,$$

where the last inequality uses $x_i \ge v(\{i\}) = 0$. □

Put $L = \{a, b\}$ and label the displacement in Theorem 3 so that $\Delta_a < 0 < \Delta_b$. By (3d), some lifted extreme packet has $r_b < 1$.

Lemma 11 (proper-hyperedge obstruction)**.** *Every such underloaded packet contains a proper rich trace R with*

$$3 \le |R| \le |H| - 1. \tag{8}$$

Proof. First dispose of the full-trace packet, for which the assertion would be vacuous. If H is itself a supported trace then $\{H\}$ is already balanced on H, so minimality forces the packet to be $\{H\}$ and there is no proper trace at all. This cannot happen at an underloaded packet. Indeed the single lifted coalition T is proper with $H \subseteq T$, hence $T \in \{H,\ H \cup \{a\},\ H \cup \{b\}\}$ and the poor load is $r \in \{(0,0), (1,0), (0,1)\}$. Underloading gives $r_b < 1$, so $r_b = 0$, and (3c) becomes $-s + r_a \Delta_a \ge 0$. That fails for $r_a = 0$ because $s > 0$, and fails for $r_a = 1$ because $\Delta_a < 0$. Every supported trace of the selected packet is therefore a proper subset of H.

Now suppose that every such trace has size at most two. The positive incidence columns of a vertex of (2) are independent. Its rank-two components are therefore isolated singletons of weight one, isolated edges of weight one, and vertex-disjoint odd cycles whose edges have weight $1/2$. This is the classical description of the vertices of a fractional perfect matching polytope, and follows by leaf deletion. In a component, independence forces at most as many positive traces as vertices. A degree-one vertex j lies in a unique trace R, and the balance equation at j forces $\beta_R = 1$; the

balance equations at the other endpoint of R then kill every remaining trace there, so R splits off as an isolated edge, or as an isolated singleton when $R = \{j\}$. Iterating removes all leaves, so a positive singleton cannot persist inside a larger component and every surviving component is a single cycle. An even cycle has dependent incidence columns, hence is excluded; an odd cycle has nonsingular incidence matrix whose unique balanced solution is the uniform weight $1/2$.

An integral packet has integer b-load, and (3c) forces that load to be at least one. Hence the selected packet is fractional. Its poor loads are half-integers. From

$$-s + r_a\Delta_a + r_b\Delta_b \geq 0, \qquad \Delta_a + \Delta_b = s,$$

and $r_b < 1$, one obtains $r_b = 1/2$ and $r_a = 0$. Exactly one half-weight edge on one odd cycle therefore carries b, and no supported coalition carries a.

Round that labeled odd cycle alternately, starting with the labeled edge, and leave every other component at its original weight. The resulting nonnegative combination of proper ϕ-tight coalitions has rich load $\mathbf{1}_H + e_d$ for one rich player d, poor load e_b, and no load on a. Exactify singleton d. Summing the selected tight inequalities and adding the omitted singleton a gives the exact-face overload inequality $\phi_d \leq \phi_a$, contradicting the strict boundary. Indeed, Lemma 10 has $D = \{d\}$, $c_b = 1$, and $c_a = 0$. Thus (8) holds. □

Proposition 12. *If $m = 2$ and the suffix is slack, then $h \geq 4$.*

Proof. Lemma 11 requires a proper trace of size at least three. □

The proof so far yields every forbidden pair in (1) without enumerating games or payoff-stratum patterns.

4 Three universal realizing families

We use a standard lower-envelope observation.

Lemma 13 (lower-envelope realization)**.** *Let G be a finite family of nonnegative allocations with common total T, and define $v(S) = \min_{g\in G} g(S)$. Then every $g \in G$ belongs to* Core(v)*, every coalition attains its worth, and v is exact. If every coordinate is zero in some row, then v is singleton-normalized.*

Proof. The defining inequality gives $g(S) \geq v(S)$ for every g, S, so every row is a core allocation. Conversely every coalition minimum is attained at a row. The last assertion is immediate. □

We record the families compactly. The displayed face inequalities give short progressive leximin proofs; no search is involved.

Proposition 14 (unique-richest family)**.** *For every $m \geq 4$ there is an integral indecomposable exact game with m poorer players $P = \{1, \ldots, m\}$, one richer player r, and*

$$\mathrm{Lmin}(v) = \mathbf{1}_P + (m-2)e_r, \qquad v(\{r\}) = 0.$$

Proof. Apply Lemma 13 to the four common-total rows

$$\begin{aligned} g^1 &= (m-1)(e_4 + e_r),\\ g^2 &= (m-1)(e_3 + e_4),\\ g^3 &= e_1 + e_2 + e_3 + (m-4)e_5 + (m-1)e_r,\\ g^4 &= \mathbf{1}_P + (m-2)e_r, \end{aligned}$$

where the zero e_5 term is omitted at $m = 4$. Every coordinate vanishes in a row. The four attained faces

$$v(N \setminus \{1\}) = v(N \setminus \{2\}) = 2m - 3, \qquad v(\{3, r\}) = v(\{4, r\}) = m - 1$$

show that any core point giving everyone at least one fixes players $1, 2$ at one; subtracting $\mathbf{1}_N$ and comparing the two pair inequalities with efficiency then fixes every poorer player at one. The remaining total gives $x_r = m - 2$. Hence $g^4 = \mathrm{Lmin}(v)$. Row g^2 proves $v(\{r\}) = 0$. For indecomposability, observe that

$$v(\{1, 4\}) = v(\{2, 4\}) = v(\{3, 4\}) = 1, \qquad v(\{3, r\}) = v(\{4, r\}) = m - 1,$$

and, when $m \geq 5$, $v(\{4, 5\}) > 0$. These positive pair worths place players $1, \ldots, 5, r$ in one direct-sum component. If $m = 6$, the coalition $T = \{3, 4, 5, r\}$ satisfies $v(T) = 7 < v(T \cup \{6\}) = 8$. If $m \geq 7$, put $T = \{4, 5\}$; then $v(T) = 2 < v(T \cup \{j\}) = 3$ for every $j \geq 6$. Since every remaining singleton has value zero, these strict interactions attach every remaining player to the same component. □

Proposition 15 (two-poorer internal-sacrifice family)**.** *For every $m \geq 2$ and $h \geq 4$ there is an integral indecomposable exact game with a poorer block P of size m, a richer suffix H of size h, and*

$$\mathrm{Lmin}(v) = D\mathbf{1}_P + 2D\mathbf{1}_H, \qquad v(H) = 2D(h - 1), \qquad D = (h - 2)(h - 3).$$

Proof. Put $r = h - 2$, write $P = \{p_1, p_2\} \dot\cup Q$ and $H = \{a, b\} \dot\cup K$, $|K| = r$. Use the common-total rows

$$\begin{aligned}
u &= De_{p_2} + D\mathbf{1}_Q + 3D(e_a + e_b) + (2r - 1)(r - 1)\mathbf{1}_K, \\
z &= 4De_{p_1} + D\mathbf{1}_Q + 2(r^2 - 1)\mathbf{1}_K, \\
w^k &= De_{p_1} + D\mathbf{1}_Q + 3D(e_a + e_b) + r(2r - 1)\mathbf{1}_{K\setminus\{k\}} \quad (k \in K), \\
\phi &= D\mathbf{1}_P + 2D\mathbf{1}_H, \\
q^j &= \phi + D(e_{p_1} - e_j) \quad (j \in Q).
\end{aligned}$$

Lemma 13 gives a normalized integral exact game. Direct substitution yields

$$\begin{aligned}
v(N \setminus \{p_2\}) &= D(m + 2h) - D, \\
v(H \setminus \{a\}) = v(H \setminus \{b\}) &= 2D(h - 1), \\
v(\{p_1, a, b\} \cup (K \setminus \{k\})) &= D(2h - 1), \\
v(H) &= 2D(h - 1).
\end{aligned}$$

At a core point x above the floor D, the complement face first fixes $x_{p_2} = D$. Write $y = x - D\mathbf{1}_N$, $E = y(Q)$, and $t = y(K)$. Then $y \geq 0$, $y_{p_2} = 0$, and $y(N) = hD$. The two co-singleton faces give

$$y_a + t \geq (h - 1)D, \qquad y_b + t \geq (h - 1)D. \tag{9}$$

The $r = h - 2$ remaining faces give, for every $k \in K$,

$$y_{p_1} + y_a + y_b + t - y_k \geq (h - 1)D.$$

Summing them and substituting $y_{p_1} + E + y_a + y_b + t = hD$ yields $t + rE \leq rD$. On the other hand, (9) and efficiency imply

$$y_{p_1} + E \leq t - rD \leq -rE.$$

Nonnegativity forces $y_{p_1} = E = 0$. Hence all of P is pinned at level D; efficiency leaves total $2Dh$ on H, equalized by ϕ. Thus $\phi = \mathrm{Lmin}(v)$ and its richer suffix has slack $2D$. The same construction is indecomposable. Indeed, p_1 has positive pair worth with every other player:

$$v(\{p_1, p_2\}) = v(\{p_1, j\}) = v(\{p_1, k\}) = D \quad (j \in Q, k \in K), \qquad v(\{p_1, a\}) = v(\{p_1, b\}) = 3D.$$

Thus the positive-pair graph is connected. □

Proposition 16 (three-poorer escape family)**.** *For every $m \geq 3$ and $h \geq 2$ there is an integral indecomposable exact game with m cumulative poorer players and richer suffix H, $|H| = h$, such that*

$$\mathrm{Lmin}(v) = \mathbf{1}_{\{p_1,p_2\}\cup Q} + 2e_{p_3} + 3\mathbf{1}_H, \qquad v(H) = 3h - 1, \qquad |Q| = m - 3.$$

Proof. Write $H = \{r_1, \ldots, r_h\}$ and take the lower envelope of

$$\begin{aligned}
a &= 2e_{p_1} + 3e_{p_3} + \mathbf{1}_Q + 5e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1,r_h\}},\\
b &= 2e_{p_2} + e_{p_3} + \mathbf{1}_Q + 4e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1\}},\\
c &= \mathbf{1}_Q + (3h+4)e_{r_h},\\
\phi &= e_{p_1} + e_{p_2} + 2e_{p_3} + \mathbf{1}_Q + 3\mathbf{1}_H,\\
q^j &= \phi + e_{p_1} - e_j \quad (j \in Q).
\end{aligned}$$

All rows have total $m + 3h + 1$, and every coordinate vanishes in a row, so Lemma 13 applies. The attained values

$$\begin{aligned}
v(N \setminus \{j\}) &= m + 3h \quad (j \in Q),\\
v(H \cup \{p_3\}) &= 3h + 2,\\
v(H \cup \{p_1\}) &= 3h + 1,\\
v(H) &= 3h - 1
\end{aligned}$$

give the progressive proof. On the face where every player receives at least one, the complement faces fix Q, the second face fixes p_1, p_2 at one, and the third fixes p_3 at two. Efficiency leaves total $3h$ on H, equalized at three by ϕ. Thus $\phi = \mathrm{Lmin}(v)$ and $s_\phi(H) = 1$. When $Q \neq \emptyset$, one member of Q has positive pair worth with every other player. When $Q = \emptyset$, the positive-pair graph connects all but p_2, and

$$v(\{p_1, p_3, r_h\}) = 4 < 5 = v(\{p_1, p_2, p_3, r_h\})$$

attaches p_2. Hence the game is indecomposable. □

Proof of Theorem 1. Corollary 7 excludes $m = 1$ for every h. Proposition 9 excludes $m \leq 3$ when $h = 1$, and Proposition 14 realizes every $m \geq 4$. Proposition 12 excludes $h \leq 3$ when $m = 2$, and Proposition 15 realizes every $m \geq 2, h \geq 4$. All remaining pairs have $m \geq 3, h \geq 2$ and are realized by Proposition 16. □

5 The four/five threshold and its mechanism

Corollary 17 (sharp player threshold)**.** *Every singleton-normalized exact game on at most four players is exact partition at* Lmin*. Five players suffice for failure.*

Proof. Every pair (m, h) with $m + h \leq 4$ is forbidden by (1). The pair $(m, h) = (3, 2)$ is realized by Proposition 16 on five players. Alternatively, $(4, 1)$ is realized by Proposition 14. □

The theorem gives two distinct first mechanisms. The $(4,1)$ obstruction is an exterior sacrifice: the fourth poorer player supplies the first positive singleton budget capable of financing a slack richest singleton. The $(3,2)$ obstruction is a mixed block-star mechanism. Two-poorer failures do not yet exist on five players: their active balanced packet must contain a proper rank-three trace, which first fits at $(2,4)$ on six players. Thus "five" is not an unexplained output of enumeration; it is the first total size at which either of the two admissible boundary regions in (1) is entered.

6 Which system the selection produces

Theorem 4 produces a system (P) but says nothing about which system. This section answers that question completely on five players and partially in general. Throughout, v is singleton-normalized and exact, $\phi = \mathrm{Lmin}(v)$, $N = L \,\dot\cup\, H$ is a strict payoff boundary with $s = s_\phi(H) > 0$, $m = |L|$ and $h = |H|$. Two conventions are used repeatedly and recorded once. Core allocations of a normalized game are nonnegative, so $\phi_i \geq 0$ for every i and hence $\phi_j > 0$ for $j \in H$; consequently no singleton $\{j\}$ with $j \in H$ is ϕ-tight, and H itself is not ϕ-tight. In (P) the coefficient of H is $-\delta < 0$ and every other coefficient is nonnegative, so H is the unique exceptional set of the system and the catalogue's constant r is the quantity b.

It is convenient to name the objects that Theorem 3 produces before Theorem 4 selects among them.

Definition 18 (normal certificate)**.** *A* normal certificate *at the boundary $L|H$ is a nonnegative weighting λ of proper ϕ-tight coalitions, each meeting H, whose incidence load equals one at every $j \in H$; writing c_i for the load at $i \in L$, $b = \max_i c_i$, $u_i = b - c_i$ and $\delta = 1 + \sum_i u_i - b$, one requires $\delta > 0$. Its* system *is*

$$\Lambda = \sum_T \lambda_T \mathbf{1}_T + \sum_{i \in L} u_i \mathbf{1}_{H \cup \{i\}} - \delta \mathbf{1}_H = b\, \mathbf{1}_N.$$

The normalized cover of Lemma 2 is a normal certificate, because its poorer loads satisfy $c_i < 1$ and therefore $\delta = F(c) \geq 1 - \max_i c_i > 0$; and the packet of Theorem 4 is the special normal certificate whose λ is a lifted vertex of $\mathcal{P}_H$. An exclusion proved for normal certificates is thus *robust*: it survives both the mixture of Theorem 3 and the Jensen selection.

6.1 Three elementary lemmas

The first is the qualitative content of the slack budget (PB).

Lemma 19 (positive load)**.** *Every normal certificate at a slack boundary has $u_i > 0$ for at least one $i \in L$. Equivalently its system carries a strictly positive coefficient on some* spoke *$H \cup \{i\}$, $i \in L$; in particular $b > 0$.*

Proof. Choose $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$ and put $\Delta = z - \phi$. Each supported T is tight and feasible, so $\Delta(T) \geq 0$; each spoke satisfies $\Delta(H \cup \{i\}) \geq -s_\phi(H \cup \{i\})$; and $\Delta(H) = -s$. Evaluating $\Lambda = b\mathbf{1}_N$ at Δ and using $\Delta(N) = 0$ gives (PB). If $u = 0$, (PB) reads $\delta s \leq 0$, contradicting $\delta > 0 < s$. Finally $u_i > 0$ forces $b > c_i \geq 0$. □

Corollary 20. *Let a permutational type have exceptional set E and no positive set of the form $E \cup \{i\}$. Then it is the type of no normal certificate at a slack boundary, hence of no packet and of no mixture. Every type with $r = 0$ has this property: if $r = 0$ and $i \notin E$ then the load at i vanishes while every set containing i has nonnegative coefficient, so every positive set lies inside E.*

The second lemma is an uncrossing inequality for tight coalitions. It was obtained independently in the companion five- and six-player computations, and is stated here once.

Lemma 21 (crossing budget)**.** *Let v be exact and singleton-normalized and let $\phi \in \mathrm{Core}(v)$. If S_1 and S_2 are ϕ-tight then*

$$s_\phi(S_1 \cap S_2) \;\le\; \phi\big(N \setminus (S_1 \cup S_2)\big).$$

Proof. Put $C = S_1 \cap S_2$, $U = S_1 \cup S_2$ and $R = N \setminus U$. Exactness supplies $z \in \mathrm{Core}(v)$ with $z(C) = v(C)$, whence

$$\phi(S_1) + \phi(S_2) = v(S_1) + v(S_2) \le z(S_1) + z(S_2) = z(U) + z(C) = \phi(N) - z(R) + v(C).$$

Since $\phi(S_1) + \phi(S_2) = \phi(U) + \phi(C)$ and $\phi(U) = \phi(N) - \phi(R)$, this is $s_\phi(C) \le \phi(R) - z(R) \le \phi(R)$, the last step by $z \ge 0$. □

Two specializations are used. If $S_1 \cup S_2 = N$ then $S_1 \cap S_2$ is tight. If $S_1 \cap S_2 = \{j\}$ then $\phi_j \le \phi(N \setminus (S_1 \cup S_2))$; taking $j \in H$ and $S_1 \cup S_2 = N \setminus \{i\}$ with $i \in L$ contradicts the strict boundary. The four-player arguments of the companion note are the case $S_1 \cup S_2 = N$.

The third lemma is the exact dual of singleton normalization and sharpens Lemma 10 at a singleton face.

Lemma 22 (zeroing obstruction)**.** *Let v be exact and singleton-normalized, $\phi \in \mathrm{Core}(v)$, and let $\lambda \ge 0$ be weights on proper ϕ-tight coalitions whose loads satisfy $c_k \le 1$ for every $k \ne j$. Then $\sum_k c_k \phi_k \le \phi(N)$; equivalently, with $c_j = 1 + \theta$,*

$$\theta\, \phi_j \;\le\; \sum_{k \ne j} (1 - c_k)\phi_k.$$

Proof. Exactness at $\{j\}$ gives $z \in \mathrm{Core}(v)$ with $z_j = v(\{j\}) = 0$. Then $\sum_T \lambda_T \phi(T) = \sum_T \lambda_T v(T) \le \sum_T \lambda_T z(T) = \sum_k c_k z_k = \sum_{k \ne j} c_k z_k \le \sum_{k \ne j} z_k = \phi(N)$, and the left side is $\sum_k c_k \phi_k$. □

Corollary 23. *If j is a richest player and $\sum_T \lambda_T |T| \ge |N|$, then $\phi_k = \phi_j$ for every k with $c_k < 1$. In particular the configuration is impossible as soon as some k with $c_k < 1$ has $\phi_k < \phi_j$.*

By linear-programming duality, Lemma 22 is not merely necessary: a core allocation vanishing at j fails to exist precisely when some such λ has $\sum_k c_k \phi_k > \phi(N)$. It is the only obstruction that is ever active in the classification below.

6.2 Full-trace packets are stars, and stars are facets

Definition 24 (star system)**.** *For $\emptyset \ne H \subsetneq N$, $L = N \setminus H$ and $\emptyset \ne Q \subsetneq L$, put*

$$\mathrm{St}(H, Q): \qquad \mathbf{1}_{H \cup Q} + \sum_{i \in L \setminus Q} \mathbf{1}_{H \cup \{i\}} - (|L| - |Q|)\, \mathbf{1}_H = \mathbf{1}_N.$$

Lemma 25 (star normal form)**.** *Let the boundary be slack.*

(i) If $h = 1$ then $\mathcal{P}_H$ has the single vertex $\beta_H = 1$, so every packet has exactly one lift T; T is proper and $T \ne H$, hence $T = H \cup Q$ with $\emptyset \ne Q \subsetneq L$, and the system is $\mathrm{St}(H, Q)$, with $b = 1$ and $\delta = m - |Q|$.

(ii) For arbitrary h, every full-trace *packet has one lift $T \supseteq H$, again proper and $\ne H$, so its system is $\mathrm{St}(H, T \cap L)$.*

In either case every positive coefficient equals one and $b = 1$.

Proposition 26 (stars are facets)**.** $\mathrm{St}(H,Q)$ *is a minimal semi-balanced system with exceptional set* H*, and it is indecomposable. Its permutational type depends only on* $(h,q)=(|H|,|Q|)$*, and the admissible pairs are* $h\geq 1$*,* $q\geq 1$*,* $h+q\leq n-1$*, so there are* $\binom{n-1}{2}$ *star types on* n *players.*

Proof. Write $\mathcal{S}=\{H\}\cup\{H\cup\{i\}:i\in L\setminus Q\}\cup\{H\cup Q\}$, $m=|L|$, $q=|Q|$, and note $1\leq q\leq m-1$, so $L\setminus Q\neq\emptyset$ and $H\cup Q\neq N$. For $k\in H$ every member of $\mathcal{S}$ contains k; for $k\in Q$ the only member containing k is $H\cup Q$; for $k\in L\setminus Q$ the only member containing k is $H\cup\{k\}$. The displayed coefficients are all nonzero with exactly one negative and give load one everywhere, so the system is semi-balanced with $r=1$.

Minimality. Let $\emptyset\neq\mathcal{S}'\subsetneq\mathcal{S}$ carry nonzero coefficients μ, at most one negative, with constant load r'. Reading the load at a player of Q and at a player of $L\setminus Q$ shows that $\mu_{H\cup Q}=r'$ if $H\cup Q\in\mathcal{S}'$ and $r'=0$ otherwise, and likewise $\mu_{H\cup\{i\}}=r'$ if $H\cup\{i\}\in\mathcal{S}'$ and $r'=0$ otherwise. If $r'=0$ then no set other than H can lie in $\mathcal{S}'$ without a vanishing coefficient, so $\mathcal{S}'=\{H\}$ and the load at H gives $\mu_H=0$, excluded. If $r'\neq 0$ then $H\cup Q$ and every $H\cup\{i\}$ lies in $\mathcal{S}'$ with coefficient r', so $\mathcal{S}'=\mathcal{S}\setminus\{H\}$ by properness, and the load at a player of H gives $(m-q+1)r'=r'$, i.e. $r'=0$ since $q<m$. Contradiction.

Indecomposability. Suppose E is a proper nonempty set with $E\notin\mathcal{S}$ and there are $\sigma\geq 0$ on $\mathcal{S}$ and $t>0$ with $\sum_{S\in\mathcal{S}}\sigma_S\mathbf{1}_S-t\mathbf{1}_E=r\mathbf{1}_N$. Write $x=\sigma_H$, $y_i=\sigma_{H\cup\{i\}}$, $z=\sigma_{H\cup Q}$, $\Sigma=x+\sum_i y_i+z$ and $\varepsilon_k=[k\in E]$. The three families of load equations read

$$\Sigma-t\varepsilon_k=r\ (k\in H),\qquad z-t\varepsilon_k=r\ (k\in Q),\qquad y_k-t\varepsilon_k=r\ (k\in L\setminus Q).$$

The first shows ε is constant on H, and the second that it is constant on Q; so $E\cap H\in\{\emptyset,H\}$ and $E\cap Q\in\{\emptyset,Q\}$.

If $E\cap H=\emptyset$ then $r=\Sigma$. For $k\in Q$ we get $z-t\varepsilon_k=\Sigma\geq z$, so $\varepsilon_k=0$ and $z=\Sigma$, whence $x=0$ and every $y_i=0$. Then for $k\in L\setminus Q$, $-t\varepsilon_k=r=z$; positivity of t forces $z=0$ and $\varepsilon_k=0$, so $E=\emptyset$, a contradiction.

If $H\subseteq E$ then $r=\Sigma-t$. Suppose first $Q\subseteq E$. Then $z=\Sigma$, so $x=0$ and every $y_i=0$; for $k\in L\setminus Q$, $-t\varepsilon_k=z-t$. If some $\varepsilon_k=1$ then $z=0$, all σ vanish and $-t\mathbf{1}_E=r\mathbf{1}_N$ forces $E\in\{\emptyset,N\}$, excluded; so every $\varepsilon_k=0$ and $E=H\cup Q\in\mathcal{S}$, excluded. Suppose instead $Q\cap E=\emptyset$. Then $z=\Sigma-t$, i.e. $x+\sum_i y_i=t>0$. If $\varepsilon_k=1$ for some $k\in L\setminus Q$ then $y_k=\Sigma$, so all other weights vanish, hence $\varepsilon_i=0$ and $y_i=r=0$ for $i\neq k$, and $E=H\cup\{k\}\in\mathcal{S}$, excluded. If $\varepsilon_k=0$ for every $k\in L\setminus Q$ then $E=H\in\mathcal{S}$, excluded.

The remaining assertions are clear: relabelling identifies any two stars with the same (h,q), and $q\leq m-1=n-h-1$. □

6.3 The five-player classification

At $n=5$ Theorem 1 admits only $(m,h)=(4,1)$ and $(3,2)$. For those two shapes one may enumerate, with no game and no payoffs, every system (P) with $\delta>0$ that Theorem 4 can produce: the traces range over the minimal balanced collections on H and each lift over every proper set with the prescribed trace that is not H and not a singleton inside H. There are three types at $(4,1)$, all full-trace, and eight at $(3,2)$, of which two are full-trace and six are split.

Theorem 27 (five-player selection)**.** *Let* v *be a singleton-normalized exact five-player game with a slack strict boundary* $L|H$ *at* $\phi=\mathrm{Lmin}(v)$*. Then a system (P) is a facet of the cone of exact games—equivalently an indecomposable minimal semi-balanced system, equivalently a member of the catalogue—if and only if its packet is full-trace, in which case it is the star* $\mathrm{St}(H,T\cap L)$*. The*

facet-defining types that occur are exactly

$$(h,q)=(1,1)\to \textit{row } 16,\quad (1,2)\to \textit{row } 14,\quad (1,3)\to \textit{row } 11,$$
$$(h,q)=(2,1)\to \textit{row } 13,\quad (2,2)\to \textit{row } 10,$$

and all five are realized by integral exact games. Systems (P) that are not facets also occur; see Example 29.

The eleven excluded rows fall into three groups of visibly different depth.

- Rows 1–8 and row 12 are excluded by Corollary 20: none of them carries a positive spoke over its exceptional set. Rows 1–8 have $r=0$; row 12 is $-a+abc+ade$, whose exceptional set is $\{a\}$ and which has no positive set of size two containing a. This exclusion is robust: those types are the systems of no normal certificate, hence of no mixture either.
- Row 9 is excluded by Theorem 1 alone. It is the star with $(h,q)=(3,1)$, and as such is a perfectly legitimate packet shape; it needs $(m,h)=(2,3)$, which $m_{\min}(3)=3$ forbids on five players.
- Row 15 is excluded by Lemma 25(i): an $h=1$ packet has $b=1$ and unit positive coefficients, whereas row 15 has $r=2$ and a coefficient 2.

Remark 28 (what the selection, rather than the geometry, excludes)**.** *The exclusions of rows* 12 *and* 15 *are both statements about systems (P), and both are correct, but they are not equally deep. Row* 15 *is the system of a normal certificate: at $h=1$ every ϕ-tight coalition has trace H, so a cover may support several lifts of the one trace, and $\lambda_{acd}=\lambda_{ace}=\lambda_{ade}=\frac{1}{3}$ has rich load one and poorer loads $(0,\frac{2}{3},\frac{2}{3},\frac{2}{3})$, all below one, giving $b=\frac{2}{3}$, $u_b=\frac{2}{3}$, $\delta=1$ and exactly the row-*15 *system. What the one-packet theorem then does is replace that cover by one of its three lifts, reporting a row-*14 *star. Row* 12 *admits no such certificate, by Corollary 20. In short: row* 12 *is excluded by the geometry, row* 15 *only by the selection.*

6.4 Split packets, and a threshold in the number of poorer players

The full-trace conclusion does *not* extend to split packets, even on five players.

Example 29 (a split packet with $\delta>0$)**.** *Apply Lemma 13 to the five common-total rows*

$$(0,1,0,3,3),\quad (1,0,0,3,3),\quad (1,1,1,2,2),\quad (2,2,0,0,3),\quad (2,2,0,3,0),$$

equivalently take the core polytope $\{x\ge 0:\ x(N)=7,\ x(abe)\ge 4,\ x(abd)\ge 4,\ x(ade)\ge 5,\ x(bde)\ge 5\}$. The game is exact and singleton-normalized, $\mathrm{Lmin}(v)=(1,1,1,2,2)$*, the strict boundary is $L=\{a,b,c\}$, $H=\{d,e\}$, and $v(H)=3<4=\phi(H)$, so $s=1$. The proper ϕ-tight coalitions are abd, abe, ade, bde, abde, whose traces on H are $\{d\}$, $\{e\}$ and $\{d,e\}$:* both *proper traces occur. The split vertex lifted by $T_1=abd$ and $T_2=abe$ has $r=(2,2,0)$, $b=2$, $u_c=2$ and $\delta=1>0$, with system*

$$\mathbf{1}_{abd}+\mathbf{1}_{abe}+2\,\mathbf{1}_{cde}-\mathbf{1}_{de}=2\,\mathbf{1}_N.$$

This system is minimal semi-balanced but decomposable, by $E=\{d\}$: $\mathbf{1}_{abd}+\mathbf{1}_{cde}-\mathbf{1}_d=\mathbf{1}_N$. The same boundary also carries the full-trace packets of types row 13 *and row* 10*.*

Split packets are nevertheless invisible to the catalogue as long as the poorer side is small. The next statement is a finite check on set systems: it enumerates every abstract system (P) with $\delta>0$ for the stated (m,h) and classifies each permutational type, and therefore applies at every boundary of every exact game.

By Proposition 12 an $h=2$ boundary needs $m\ge 3$, so the statement below starts where Lemma 11 stops.

Theorem 30 (split packets and the poorer-side threshold)**.** *Let $h = 2$. If $m \le 4$ then every split packet system is minimal but decomposable; consequently every facet-defining packet at an $h = 2$ boundary with $m \le 4$ is a full-trace star. This fails at $m = 5$. Precisely:*

	packets	*full-trace types*	*split types*	*split facets*
$m = 3$ $(n = 5)$	*42*	*2*	*6*	*0*
$m = 4$ $(n = 6)$	*216*	*3*	*13*	*0*
$m = 5$ $(n = 7)$	*950*	*4*	*24*	*2*

The two exceptions at $m = 5$ are $af - 5fg + bfg + cfg + 2dfg + 2efg + abcg$ and $-5fg + abf + acg + bfg + cfg + 2dfg + 2efg$. Equivalently, no catalogued type with a doubleton exceptional set has split-packet shape when $m \le 4$: on five players the doubleton rows are $1, 10, 13$, of which 10 and 13 are stars and 1 has no spoke; on six players they are $1, 25, 28, 29, 33, 34, 39$, of which $28, 33, 39$ are stars and none is split.

The same finite check explains the analogous phenomenon one dimension up.

Theorem 31 (which trace collections carry facets at $h = 3$)**.** *Let $(m, h) = (3, 3)$, so $n = 6$. Among the 811 abstract packets with $\delta > 0$, sorted by the minimal balanced collection their traces form:*

collection on H	*types*	*facets*	*rows*
$\{H\}$ *(full trace)*	*2*	*2*	32, 27
three pairs (the 2-cover)	*31*	*5*	7, 24, 51, 52, 53
a singleton and a pair	*10*	*0*	—
three singletons	*12*	*0*	—

Only the full trace and the two-cover carry facet-defining systems.

Theorems 30 and 31 share one mechanism: the partition-type minimal balanced collections generate decomposable systems, so they can certify a failure but never a facet of the exact cone. The five-player classification is the case where that mechanism is complete.

6.5 What is proved, what is checked, and what is verified

status	**statement**
proved	Lemma 19 and Corollary 20; Lemma 21; Lemma 22 and Corollary 23; Lemma 25; Proposition 26; the exclusion of rows 1–8, 9, 12, 15 on five players; Remark 28.
finite check	the list of eleven abstract packet shapes on five players; Theorem 30; Theorem 31. Each enumerates set systems only, so each holds at every boundary of every exact game.
verified	Example 29 and the five realizing games of Theorem 27, by exact rational recomputation of the envelope, of Lmin, of the tight family and of every packet. That exactly one of the six split shapes on five players is realizable rests on a search over tight families of at most $\lvert N\rvert+2$ coalitions, which is exhaustive when the poorer side is a single payoff stratum and a broad search otherwise.

Two caveats are owed. First, the claim that the split shape of Example 29 is the *only* realizable one is the one statement above that is not a theorem; the five excluded shapes survive the tangent-cone conditions of Lemma 2 and are killed only by singleton normalization, i.e. by Lemma 22, and we have no human proof of that. Second, earlier random harvests reported no split packets at all on five players; Example 29 shows that this was a sampling artefact, and the same caution applies to any statement of the form "collection $\mathcal{R}$ is never selected" that rests on random exact games rather than on a finite check.

7 What the theorem does and does not establish

The normal-cover and one-packet theorems are dimension-free: every failure is reduced to a single minimal balanced rich-trace packet and its poorer-side completion budget. The phase theorem then classifies every possible pair of boundary-side cardinalities, and Section 6 classifies the selected systems completely on five players and by finite check for $(m, h) \leq (4, 2)$ and $(3, 3)$. What remains unclassified is the general dimension: distinct irreducible packet classes occur (full-trace stars, the two-cover class of Theorem 31, and from $m = 5$ on, facet-defining split classes), so a complete catalogue is multi-class, not a theorem that every failure contains one fixed finite configuration.

One calibration is owed to the reader. The linear-size bound in Corollary 6(i) is not itself new: it is the standard fact that a basic solution of a system of equalities has support at most the number of rows, applied to (2), whose rows are indexed by H. Sparse supports of this kind are routine in the balancedness literature surrounding the Kohlberg criterion. What is claimed here is not the bound but which single system the Lmin geometry selects, and on which prescribed suffix it lives.

The theorem's novelty remains conditional on specialist review. Exact partition, its Lorenz consequence, property I, semi-balanced inequalities, the semi-balanced/minimal-balanced correspondence, minimal balanced collections, and lower-envelope exactification are not claimed. The candidate new content is the simultaneous Lmin normal cover, the richer-side one-packet selection with its linear-size witness and slack-propagation corollary, the sharp staircase (1), the three dimension-free indecomposable realizing families, and the selection results of Section 6: the positive-load, crossing-budget and zeroing lemmas, the star normal form with its facet proposition, the five-player selection theorem, and the poorer-side threshold for split facets.

Reproducibility note

The proof above is human and enumerates neither games nor payoff-stratum patterns. Exact rational programs in the companion repository independently reconstruct the smallest family members, all coalition values, and staged Lmin solutions, and classify the packet systems of Remark 5 against the published catalogue of semi-balanced types. They are bug audits, not proof evidence.

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